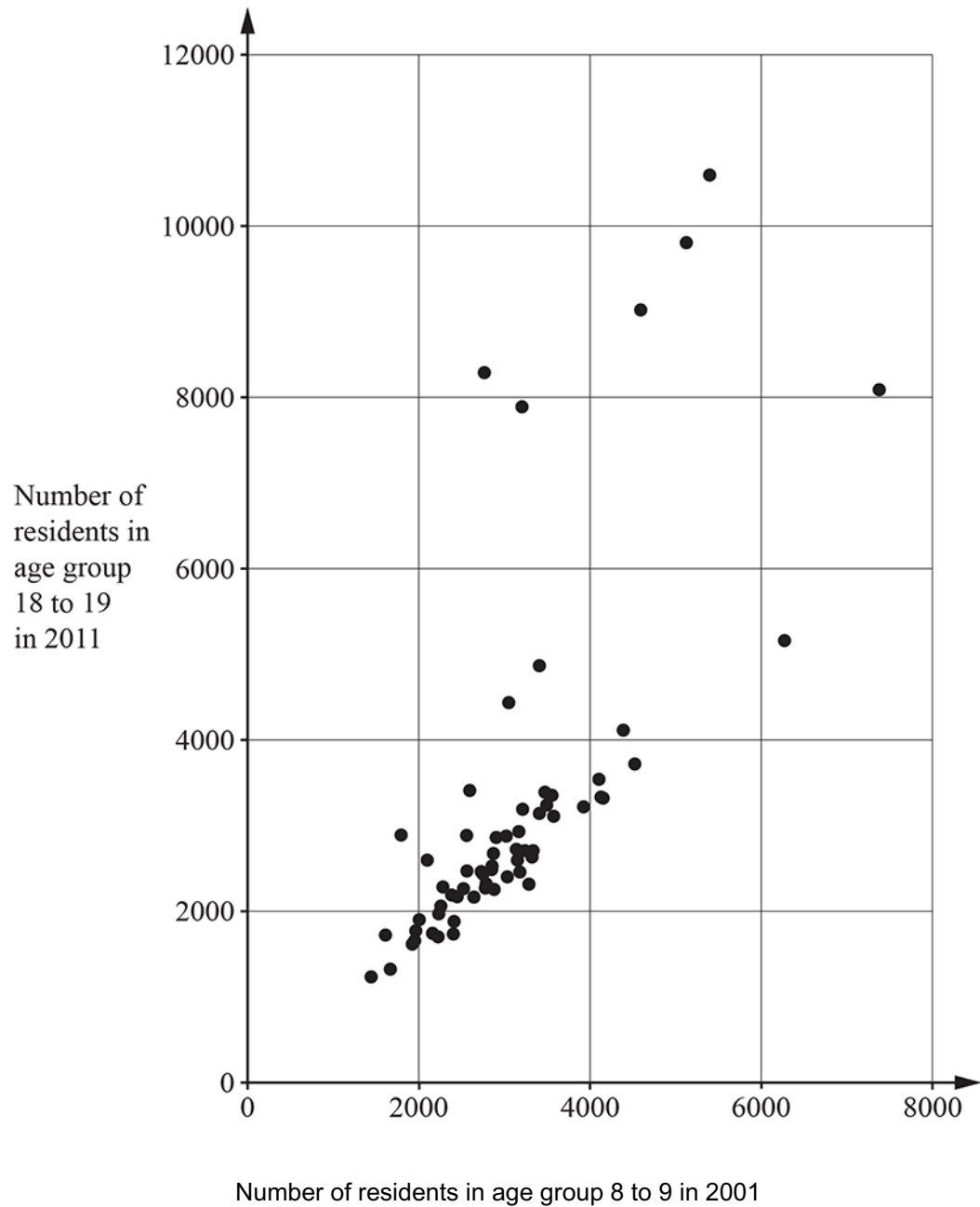


- 1(a) A student is investigating changes in the number of residents in Local Authorities in the South-East Region between 2001 and 2011. The scatter diagram shows the number x of residents in these Local Authorities in the age group 8 to 9 in 2001 and the number y of residents in the same Local Authorities in the age group 18 to 19 in 2011.



Suggest a reason why the student is comparing these two age groups in 2001 and 2011.

[1]

(b) The student notices that most of the data points are close to the line $y = x$.

(i) Explain what this suggests about the residents in these Local Authorities.

[1]

(ii) The student says that correlation does not imply causation, so there is no causal link between the values of x and the values of y . Explain whether or not they are correct.

[1]

(c) Some of these Local Authorities contain universities.

(i) On the diagram, circle three points that are likely to represent Local Authorities containing universities.

[1]

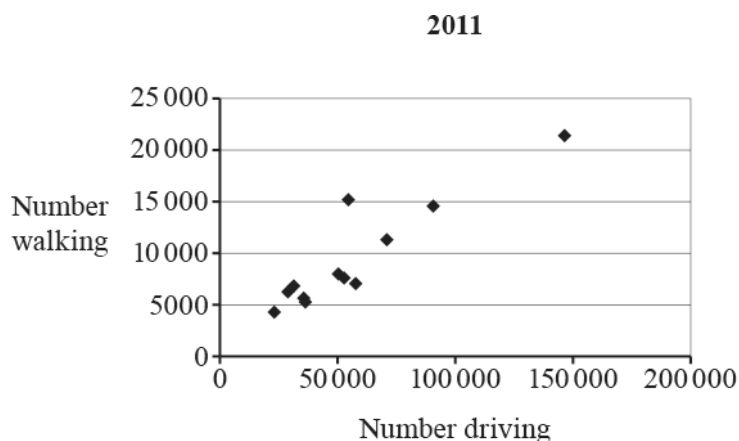
(ii) Give a reason for your choice of points in part (i).

[1]

(d) Assume that the proportion of residents in age group 8 to 9 in 2001 was roughly the same in each Local Authority in the South-East. The Local Authority in this region with the largest population is Medway. On the diagram, label clearly with the letter M the point that corresponds to Medway.

[1]

- 2 The scatter diagram shows data, taken from the pre-release data set (see <http://www.ocr.org.uk/Images/308727-units-h230-and-h240-large-data-set-lds-sample-assessment-material.xlsx>), for several Local Authorities in one region of the UK in 2011. The diagram shows, for each Local Authority, the number of workers who drove to work, and the number of workers who walked to work.



- (a) Four students calculated the value of Pearson's product-moment correlation coefficient for the data in the diagram. Their answers were 0.913, 0.124, -0.913 and -0.124 . One of these values is correct. Without calculation state, with a reason, which is the correct value. [2]

- (b) Sanjay makes the following statement.

"The diagram shows that, in **any** Local Authority, if there are a large number of people who drive to work there will be a large number who walk to work."

Give a reason why this statement is incorrect. [2]

(c) Rosie makes the following statement.

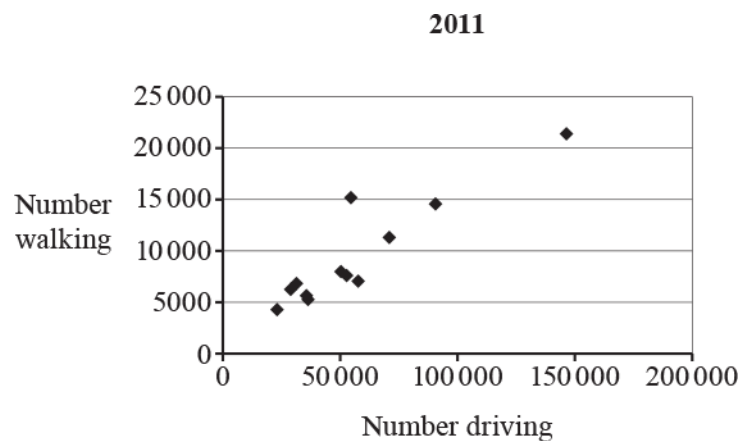
“The diagram must be wrong because it shows good positive correlation. If there are more people driving to work, there will be fewer people walking to work, so there would be negative correlation.”

Explain briefly why Rosie’s statement is incorrect.

[1]

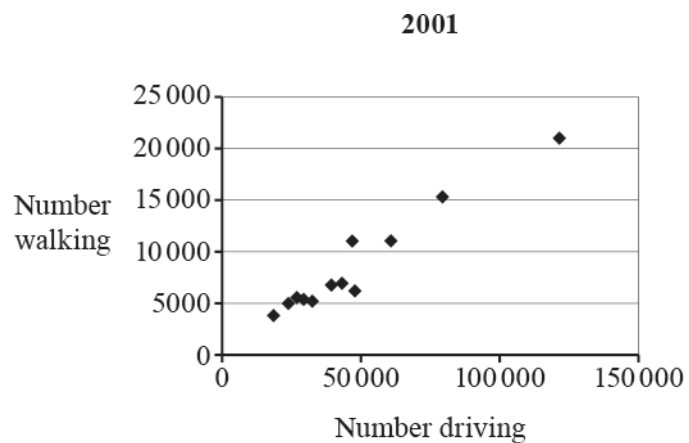
(d) The diagram shows a fairly close relationship between the two variables. One point on the diagram represents a Local Authority where this relationship is less strong than for the others. On the diagram below, label this point A.

[1]



- (e) Given that the point A represents a metropolitan borough, suggest a reason why the relationship is less strong for this Local Authority than for the others in the region. [1]

The scatter diagram below shows the corresponding data for the same region in 2001.

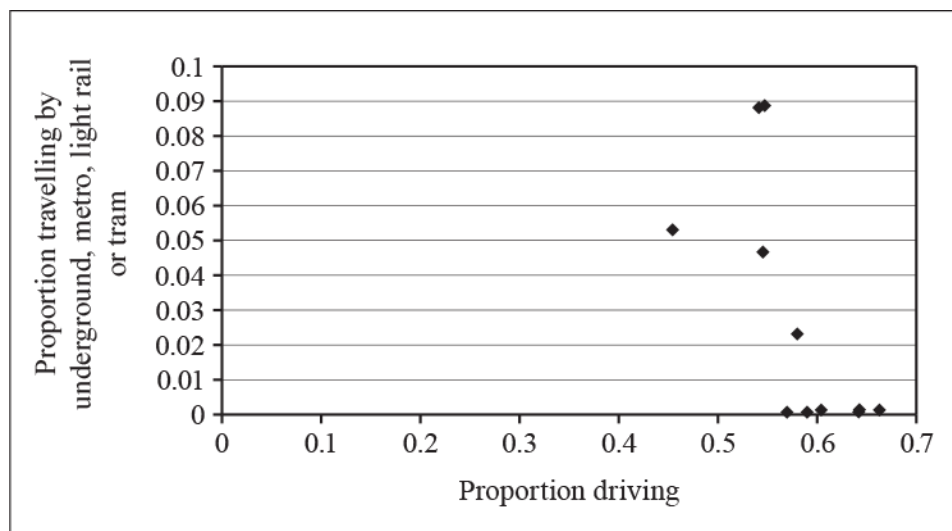


- (f) (i) State a change that has taken place in the metropolitan borough represented by the point A between 2001 and 2011. [1]

(ii) Suggest a possible reason for this change.

[1]

- 3 The scatter diagram shows data taken from the pre-release data set for several Local Authorities in a region of the UK. The diagram shows, for each Local Authority, the proportion of workers driving to work, and the proportion travelling to work by underground, metro, light rail or tram.



- (a) On the diagram above, identify the points corresponding to two distinct sections of the population represented in the diagram. [1]

- (b) Suggest a reason why there are two distinct sections of the population represented by the points in the diagram. [1]

The data for another local authority in this region can be represented by the point (0.62, 0.004).

- (c) (i) To which of the two distinct sections of the population does this Local Authority belong? Explain your answer. [1]

- (ii) What can you deduce about this Local Authority? [1]

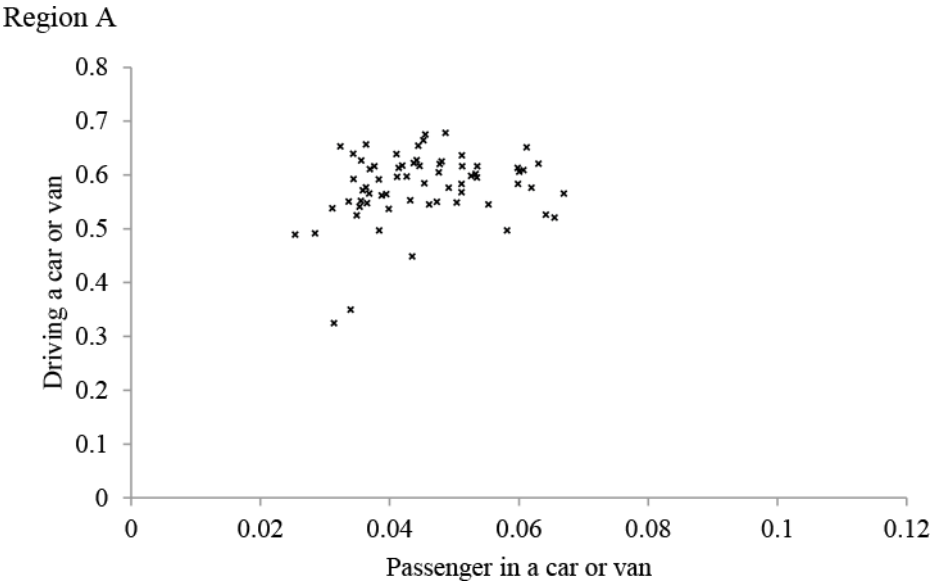
- (d) A student suggests that the Local Authority represented by the point (0.55, 0.089) is a non-metropolitan district. Comment on this suggestion. [1]

4 The table and the four scatter diagrams below show data taken from the 2011 UK census for four regions. On the scatter diagrams the names have been replaced by letters.

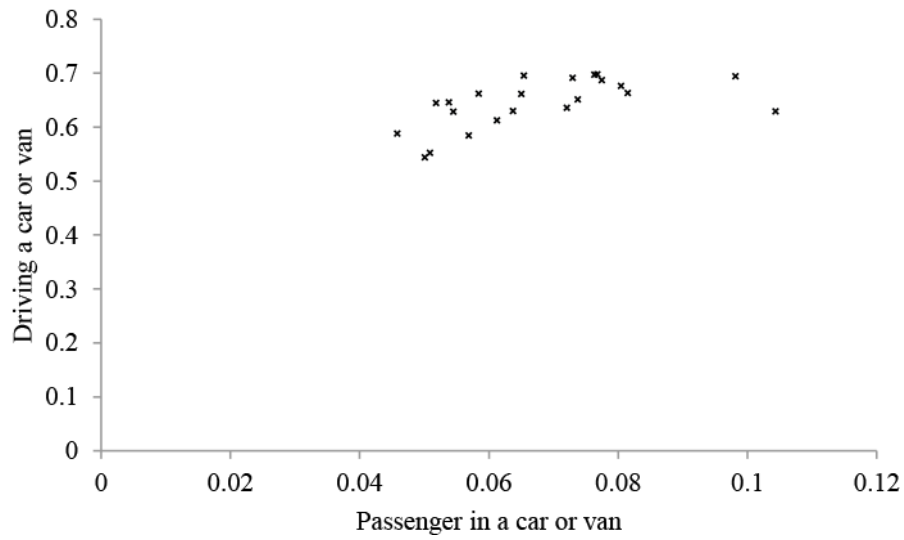
The table shows, for each region, the mean and standard deviation of the proportion of workers in each Local Authority who travel to work by *driving* a car or van and the proportion of workers in each Local Authority who travel to work as a *passenger* in a car or van.

Each scatter diagram shows, for each of the Local Authorities in a particular region, the proportion of workers who travel to work by *driving* a car or van and the proportion of workers who travel to work as a *passenger* in a car or van.

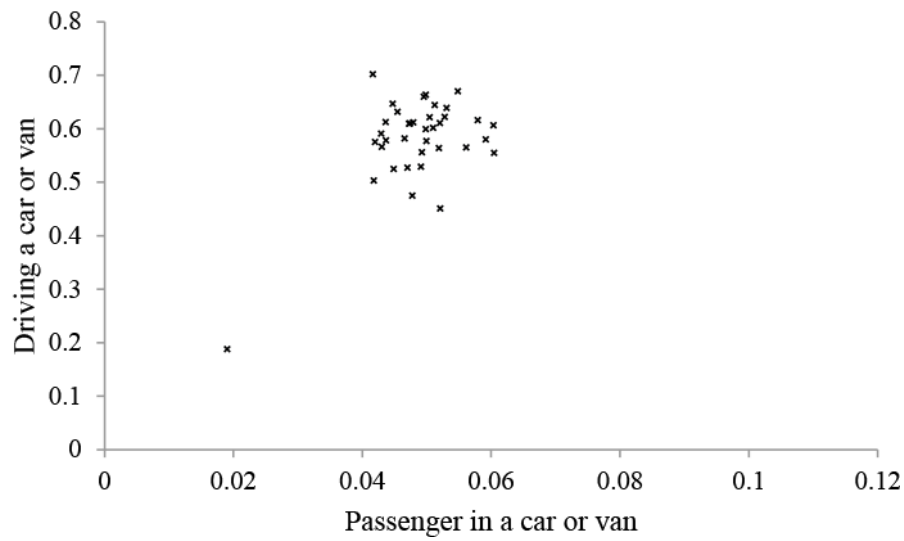
	Driving a car or van		Passenger in a car or van	
	Mean	Standard deviation	Mean	Standard deviation
London	0.257	0.133	0.017	0.008
South East	0.578	0.064	0.045	0.010
South West	0.580	0.084	0.049	0.007
Wales	0.644	0.045	0.068	0.015



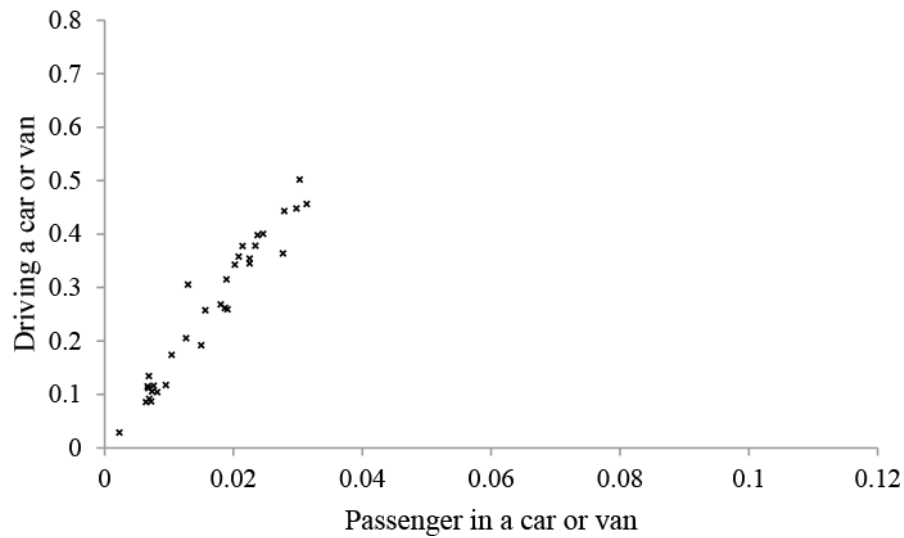
Region B



Region C



Region D



- (a) Using the values given in the table, match each region to its corresponding scatter diagram, explaining your reasoning.

[3]

- (b) Steven claims that the outlier in the scatter diagram for Region C consists of a group of small islands. Explain whether or not the data given above support his claim.

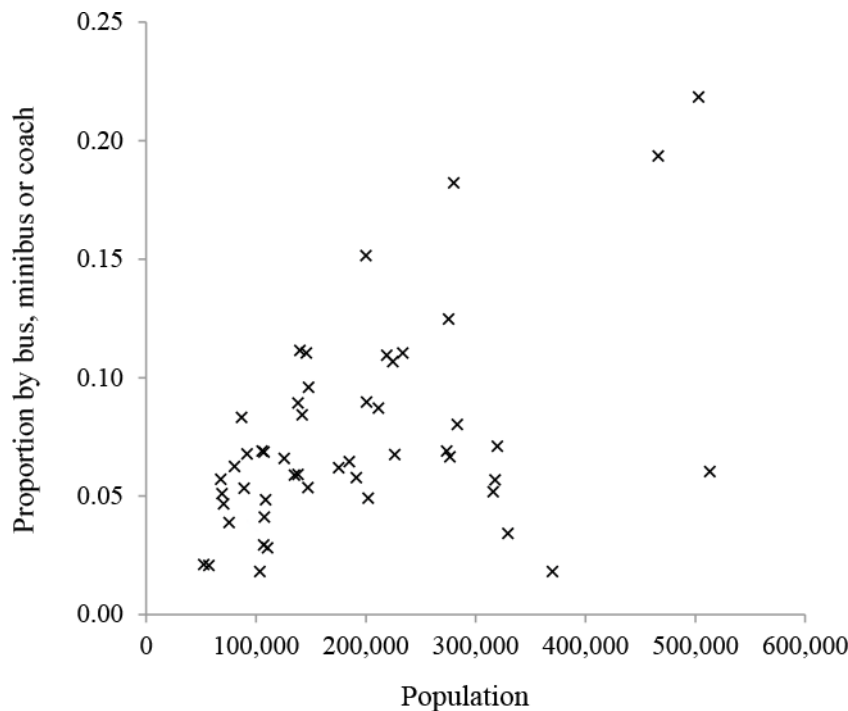
[1]

- (c) One of the Local Authorities in Region B consists of a single large island. Explain whether or not you would expect this Local Authority to appear as an outlier in the scatter diagram for Region B.

[1]

- 5 The scatter diagram below shows data taken from the 2011 UK census for each of the Local Authorities in the North East and North West regions.

The scatter diagram shows the total population of the Local Authority and the proportion of its workforce that travel to work by bus, minibus or coach.



- (a) Samuel suggests that, with a few exceptions, the data points in the diagram show that Local Authorities with larger populations generally have higher proportions of workers travelling by bus, minibus or coach. On the diagram above draw a ring around each of the data points that Samuel might regard as an exception.

[1]

- (b) Jasper suggests that it is possible to separate these Local Authorities into more than one group with different relationships between population and proportion travelling to work by bus, minibus or coach. Discuss Jasper's suggestion, referring to the data and to how differences between the Local Authorities could explain the patterns seen in the diagram.

[3]

6(a) Zac is planning to write a report on the music preferences of the students at his college. There is a large number of students at the college.

State one reason why Zac might wish to obtain information from a sample of students, rather than from all the students. [1]

(b) Amaya suggests that Zac should use a sample that is stratified by school year.

Give one advantage of this method as compared with random sampling, in this context. [1]

(c) Zac decides to take a random sample of 60 students from his college. He asks each student how many hours per week, on average, they spend listening to music during term. From his results he calculates the following statistics.

Mean	Standard deviation	Median	Lower quartile	Upper quartile
21.0	4.20	20.5	18.0	22.9

Sundip tells Zac that, during term, she spends on average 30 hours per week listening to music.

Discuss briefly whether this value should be considered an outlier. [3]

- (d) Layla claims that, during term, each student spends on average 20 hours per week listening to music. Zac believes that the true figure is higher than 20 hours. He uses his results to carry out a hypothesis test at the 5% significance level.

Assume that the time spent listening to music is normally distributed with standard deviation 4.20 hours.

Carry out the test.

[7]

- 7 Frances used the pre-release data set to produce the following table which shows information about the residents of Norwich in 2011.

Age	0 to 15	16 to 24	25 to 44	45 to 64	65 and over	Total
Number of residents	21 707	22 921	40 894	27 645	19 345	132 512

- (a) State the upper class boundary of the “25 to 44” class. [1]

Frances used these data to calculate estimates of the mean and standard deviation of the ages of these residents. She assumed that the oldest resident was aged 105.

- (b) Calculate these estimates. [3]

- (c) Use these estimates to discuss whether there may be any outliers. [2]

- (d) Jacob suggested that more reliable estimates could be obtained by using the value 80 for the last class, instead of the midpoint. Explain, with a reason, whether you think this suggestion is a good one. [1]

- 8 A researcher is studying changes in behaviour in travelling to work by people who live outside London, between

2001 and 2011.

He chooses the 15 Local Authorities (LAs) outside London with the largest decreases in the percentage of people driving to work, and arranges these in descending order.

The table shows the changes in percentages from 2001 to 2011 in various travel categories, for these Local Authorities.

Local Authority	Work mainly at or from home	Under-ground, metro, light rail, tram	Train	Bus, minibus or coach	Driving a car or van	Passenger in a car or van	Bicycle	On foot
Brighton and Hove	3.2	0.1	1.5	0.8	-8.2	-1.5	2.1	2.3
Cambridge	2.2	0.0	1.6	1.2	-7.4	-1.0	3.1	0.6
Elmbridge	2.9	0.4	4.1	0.2	-6.6	-0.7	0.3	-0.3
Oxford	2.0	0.0	0.6	-0.4	-5.2	-1.1	2.2	2.1
Epsom and Ewell	1.6	0.4	3.9	1.1	-5.2	-0.9	0.0	-0.6
Watford	0.7	2.0	3.1	0.4	-4.5	-1.2	0.0	-0.1
Tandridge	3.3	0.2	4.0	-0.1	-4.5	-1.1	0.0	-1.3
Mole Valley	3.3	0.1	1.9	0.3	-4.4	-0.7	0.2	-0.3
St Albans	2.3	0.3	3.4	-0.3	-4.3	-1.2	0.3	-0.2
Chiltern	2.9	1.4	1.4	0.1	-4.2	-0.6	-0.2	-0.8
Exeter	0.7	0.0	1.0	-0.6	-4.2	-1.5	1.7	3.4
Woking	2.1	0.1	3.7	0.0	-4.2	-1.3	-0.1	0.0
Reigate and Banstead	1.8	0.1	3.2	0.6	-4.1	-1.0	0.1	-0.2
Waverley	4.3	0.1	2.5	-0.5	-3.9	-0.9	-0.3	-0.9
Guildford	2.7	0.1	2.4	0.2	-3.6	-1.2	0.0	-0.3

The researcher suggested that there would be strong correlation between the decrease in the percentage driving to work and the increase in percentage working mainly at or from home.

Without calculation, use data from the table to comment briefly on this suggestion.

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		Those in age group 8 to 9 in 2001 will be in age group 18 to 19 in 2011	B1 [1]	or, eg, "Same people" May be implied
	b	i	Few people joined (or left) this group oe or "Not much change in this group" or "Most have lived there since 2001"	B1 [1]	LAs with a high number of 8-9s in 2001 have a high number of 18-19s in 2011 Most have stayed in the same area since childhood
		ii	(No) The nos of 18-19's in 2011 is (largely) caused by the nos of 8-9s in 2001, (but this cannot be inferred from the correlation).	B1 [1]	(No) There is a link. They are the same people Must state or imply that there is a link between x and y in this context.
	c	i	3 (or 4 or 5) of the highest five y-values	B1 [1]	Must not include right-most point
		ii	In these LAs, the no. of 18-19s in 2011 is more than the no. of 8-9s in 2001 In these LAs there are large numbers of 18+	B1 [1]	eg, the no. of people in this group in 2011 is more than it was in 2001 Students are aged 18+ so they cause an increase in this age group Allow numerical example. May be implied Not just "These are outliers" oe
	d		Right most point indicated	B1 [1]	
			Total	6	
2		a	Points close to straight line with +ve gradient Hence 0.913 is the correct value	B1(AO 2.4) B1(AO 2.2b) [2]	Dep on 1st B1
		b	Sample is from one area, hence not random oe	E2(AO 2.3 2.2b) [2]	Or might be different relationship elsewhere Allow E1 for "Sample is small"

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		c	Both depend on the size (of the pop) of area.	E1(AO 2.4) [1]	
		d	Correct point indicated (54200, 15300)	B1(AO 3.2b) [1]	
		e	More local jobs (so higher proportion walk)	E1(AO 2.2b) [1]	Any sensible equivalent,
		f	(i) Eg Fewer walk to work (ii) Eg Some businesses within the borough have closed down or have moved to the outskirts	E1(AO 2.2b) [1] E1(AO 2.2b) [1]	 or any relevant comment
			Total	9	

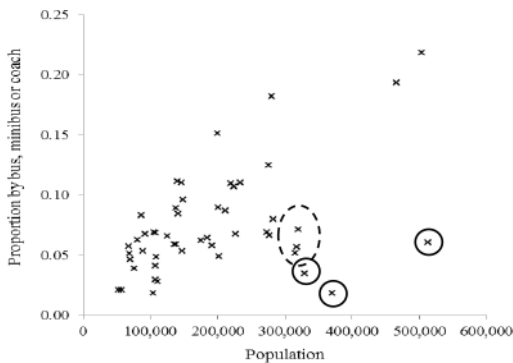
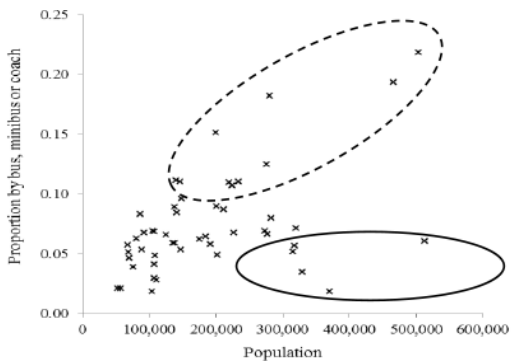
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
3		a	Identify points on (or close to) x-axis, & those not	B1(AO1.1) [1]		
		b	Some (or 5 or 6) areas have no metro etc	E1(AO2.2a) [1]	In some (or 5 or 6) areas, none use metro	Any equivalent description, in context
		c	i) Either few use metro so "no metro" group Or some use metro so in group with metro	E1(AO2.2a) [1]	or no metro in area so "no metro" group	
		c	ii) Probably travel to diff area to get metro	E1(AO2.4) [1]		
		d	Unlikely. Large prop use metro etc	E1(AO2.2b) [1]	Unlikely. There is metro etc.	
			Total	5		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
4		a	E.g. The only region with very low location on both variables is Region D which is therefore London.	E1(AO2.2a)	Or any other valid reason to connect Region D with London	OR E1 for one region correct with good reasoning
			E.g. The region with the lowest standard deviation is Region B, so this is Wales	E1(AO2.2a)	Or any other valid reason to connect Region B with Wales	OR E2 for two regions correct with good reasoning
			E.g. The only value where the other two differ much is sd of <i>driving</i> ; the wider spread on Region C including the outlier suggests that this is the Southwest, so Region A is the South East.	E1(AO2.2b)	Careful argument involving mean and / or standard deviation	
				[3]		
		b	E.g. No the data only shows that this LA has low proportions of car use for travelling to work. E.g. No, many LAs in Region D (London) have similar proportions and they are not small islands.	E1(AO2.2b)	Or any other valid explanation of why the data given is insufficient to draw this conclusion	Identifying the LA as the Scilly Isles is not relevant; this requires information that is not in the supplied data.
				[1]		
		c	E.g. On a large island, methods of travel to work are unlikely to be different to any other LA; people will still be travelling to work on the roads, and provision of public transport will be similar to any other LA.	E1(AO2.2b)	Or any other valid explanation of how large islands are likely to have similar patterns of method of travel to other LAs	Candidates may, but need not, identify the LA as Anglesey, but this is not sufficient to award the mark
				[1]		
			Total	5		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5		a		B1 (AO2.2b) [1]	At least the three with solid rings. No extras other than those in the dashed ring.
		b	 e.g. the dotted ringed group are “metropolitan districts” which have good infrastructure, so they have high proportions of travelling by bus. The solid ringed group are probably large “unitary authorities” which are not urban, so they don’t have good bus services. The unringed points are a mix of small “unitary authorities” and “non-metropolitan districts” which are difficult to tell apart with these data.	B1 (AO2.2b) E1 (AO1.2) E1 (AO2.3) [3]	For identifying (not necessarily using the diagram) the two subpopulations shown as being one in which there is a positive correlation between the two variables, and one in which larger populations do not appear to lead to increases in the proportion travelling by bus. For identifying two distinct subpopulations in terms of the structure of the large data set For explaining why it might be difficult to tell the others apart. Identifying <i>some</i> points of those ringed as being in different sub-populations
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a		Population large oe	B1 [1]	or, eg Would take too long to contact all students. NOT "Easier"
	b		eg: Includes students from all years (or ages) Numbers in years in correct proportions Different years might like different music	B1 [1]	or: Different years may have different numbers of students NOT "It's more representative" or "Takes all students into account" "You get a range of people" "It avoids bias"
	c		$21 + 2 \times 4.2 = 29.4$ $22.9 + 1.5(22.9 - 18.0) = 30.25$ Unclear whether 30 is an outlier	B1 B1 B1 [3]	Allow "30 is more than 2 sds away from the mean" Allow "30 is less than $1.5 \times \text{IQR}$ from UQ" or eg "It depends which definition you use." Any comment implying uncertainty Ignore comments about mean ± 3 sds or mean ± 1 sd <u>Examiner's Comments</u> The specification (paragraph 2.02j) gives two definitions for an outlier. Most candidates used only one of these definitions, and so did not discover that they give different results. What was required for full marks was a recognition of the uncertainty about whether or not 30 should be considered an outlier. Candidates who used other definitions (such as $x > \mu + 3s$ or $x > \mu + s$) gained no credit. Some candidates did a calculation that mixed the median with the standard deviation or the mean with the IQR.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	d	H0: $\mu = 20$ H1: $\mu > 20$ where μ = pop mean time spent $\bar{X} \sim N(20, \frac{4.2^2}{60})$ and $\bar{X} = 21$ $P(\bar{X} > 21) = 0.0326$ Compare 0.05 Alternative methods for M1A1A1 or $\frac{a-20}{4.2/\sqrt{60}} = 1.645$ (a = 20.9) CV = 20.9 21 > 20.9 or 21 not in acceptance region or $\frac{21-20}{4.2/\sqrt{60}} (= 1.84)$ zcalc = 1.84 Compare 1.645 $\bar{X} \sim N(20, \frac{4.2^2}{60})$ and $\bar{X} = \frac{1260}{60}$ or 21 $P(\bar{X} < 21) \leq 0.9674$ Compare 0.95 Reject H₀ Condon Accept H₁ There is evidence that (mean) time spent is > 20 hours or eg there is evidence to support Zac's belief NB Use of $4.2^2/60$ as sd gives $p = 0.000335$	B1 B1 M1 A1 A1 Alternative methods for M1A1A1 M1 A1 A1 M1 A1 A1 M1 A1f [7]	NB Allow 2 sf throughout Allow other letters, not X unless defined. Not $\bar{X}$ B1B0 for 1 error eg 2-tail or: 2-tail B1B0 $\mu =$ B1B0 sample mean implied undefined B1B0 Not B1B0 μ include value 20 not in B1B0 eg H₀ = 20 B1B0 terms of etc: parameter Correct distribution and value of $\bar{X}$ stated or implied eg by 0.0326 or 0.967 or 20.9 or 1.84 or 0.000335 even if within incorrect statement eg $P(X = 21) = 0.0326$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ BC Allow 2 sf, ie 0.033 Dep 0.0326 or 1.84 or 0.9674 or $P(X > 21)$ or $P(\geq 21)$ soi Must compare like with like, eg NOT prob of z-value or large prob of small prob or CV of wrong end of acceptance region Alternative methods for M1A1A1 Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2}{60}$ BC Dependent on clearly valid comparison of like with like. Dep 0.0485 or 0.951 or 20.9 or 1.84 or $P(X > 21)$ or $P(X \geq 21)$ soi May be implied by conclusion, eg "There is evidence that mean time is > 20 hours" M1A1

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					In context, not definite; eg "Mean time is > 20 hours": A0 But "There is evidence to reject H0 and that mean time is > 20 h" Allow opposite conclusion, ft their values, if above conditions met and loses 2nd A1. But potentially can score all the other 6 marks <u>Examiner's Comments</u>  Misconception One difficult feature of hypothesis testing is that in some cases the calculated value of the statistic is significant when it is greater than the critical value, whereas in other cases the calculated value is significant when it is less than the critical value. In both these questions there was confusion about this. Many candidates used the sample mean, 21, as the mean of the distribution, rather than the suggested population mean of 20.
			Total	12	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
7		a	45	B1(AO3.1b) [1]	Allow 44 years 364 days or similar
		b	$\bar{x} = 39.6$ (3 sf) $s = 24.3$ (3 sf)	B1(AO1.1) M1(AO1.1) A1(AO1.1) [3]	(UCB = 106 because oldest is 105) (If seen) sub in correct formula using any x 's within classes Allow 24.2 (3 sf) If use 105 as UCB: mean = 39.5 (3 sf) B0 sd = 24.2 (3 sf) M1A1
		c	$39.6 + 2 \times 24.3 = 88.2$ $39.6 - 2 \times 24.3 = -9$ Hence <u>may be</u> outliers at top, but not at bottom	M1(AO1.1a) A1f(AO2.2b) [2]	Allow just $39.6 + 2 \times 24.3 = 88.2$ for M1 A1 for both limits and full conclusion ft their $\bar{x}$ and s
		d	No, most in 65+ class will be nearer the lower end oe	E1(AO3.2b) [1]	or imply class is weighted towards left
			Total	7	
8			Not the case. The “driving” figures increase, but the “home” figures have no pattern.	B1(AO2.3) [1]	Not enough to quote individual LAs. Not Weak correlation
			Total	1	